Math 150 – Exam 2 – Chapters 3, 4 – Fall 2015

I pledge to demonstrate personal and academic integrity in all matters with my fellow students, the faculty, and the administration here at USD. I promise to be honest and accountable for my actions; and to uphold the stature of scholastic honesty to better myself and those around me. Initials: ___________

Name: ____ Section: **1** **2**

Instructions: **Read all directions carefully. Ask for clarification if you do not understand the directions. Show your work when necessary. Complete sentences and justifications are only necessary when indicated.** There are 6 pages to this exam.

There are 100 points on this test and 3 Extra Credit points.

Num.	Score	Out of
1		5
2		15
3		10
4		25
5		15
6		10
7		10
8		10
B		*3*
Total		100

1. Take a deep breath. Slowly write "I got this."

2. State whether each statement is True or False.

   (a) {3 points} If $y = e^3$, then $y' = 3e^2$.

   (b) {3 points} $\frac{d}{dx}(\ln 10) = \frac{1}{10}$.

   (c) {3 points} $\frac{d}{dx}(\sin(x^2 + 3x)) = \cos(2x + 3)$.

   (d) {3 points} $\cot^{-1}(x) = \frac{\cos^{-1}(x)}{\sin^{-1}(x)}$.

   (e) {3 points} $e = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n$.

3. {10 points} Write an equation of the tangent to the curve of $f(x) = \frac{x^2 - 1}{x^2 + 1}$ at the point $(-1, 0)$

4. Calculate y'. You do not need to simplify.

(a) {5 points} $y = x^4 - 3x^2 + 5$

(b) {5 points} $y = (x^4 - 3x^2 + 5)^4$

(c) {5 points} $y = \dfrac{3}{x^4}$

(d) {5 points} $y = \ln(3x^2 + e^{\cos(x)})$

(e) {5 points} $y = \tan^2(x^3)$

5. Use implicit differentiation to find $y' = \frac{dy}{dx}$. Solve for $\frac{dy}{dx}$ (or y' if that is what you are using.)

(a) {5 points} $9 = x^2 + y^2$

(b) {5 points} $\sin(y) = xy - x^2$.

(c) {5 points} $xe^y = y - 1$

6. {10 points} The mass of a part of wire is $x(1+\sqrt{x})$ kilograms, where x is measured in meters from one end of the wire. Find the linear density of the wire when $x = 4$ meters.

7. {10 points} Find the linearization $L(x)$ of the function $f(x) = \sqrt[3]{x}$ at $a = 8$.

8. Differentiate.

   (a) {5 points} $y = \sin^{-1}(x^3)$

   (b) {5 points} $y = x^{\tan(x)}$

*{3 points} * **Bonus Question: This is for extra credit** (use logarithmic differentiation).

9. $y = \sqrt[3]{\dfrac{(x^2+1)^3 x^2}{(5x+1)(x^4+2)}}$